

Mathematics 629
Spring 2026
Homework Assignment 6

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Due: 3/25/2026

1. Write out the complete details of the proof of the “differentiation under the integral” Theorem 1.129 in the notes.

Proof. i) Suppose there exists $g \in L^1(\mu)$ such that $|f(x, t)| \leq g(x)$ for all (x, t) and $\lim_{t \rightarrow t_o} f(x, t) = f(x, t_o)$. Let $f_n(x) = f(x, t_o)$ so $f_n(x) \rightarrow f(x, t)$. Since $g(x)$ dominates $f(x, t)$ and f_n converges pointwise, by DCT,

$$\lim \int f_n d\mu = \int f(x, t_o) d\mu$$

or equivalently $F(t_n) \rightarrow F(t_o)$. Since $t_n \rightarrow t_o$ is an arbitrary sequence then $\lim_{t \rightarrow t_o} F(t) = F(t_o)$ for all t_o .

ii) Suppose there also exists $t \in [a, b]$ and partial derivative $\frac{\partial f}{\partial t}$ exists such that $|\frac{\partial f}{\partial t} f(x, t)| \leq g(x)$ almost everywhere. Let $d_n(x) = \frac{f(x, t+h_n) - f(x, t_o)}{h_n}$ be a sequence where $h_n \rightarrow 0$. Then by definition of derivative, $d_n \rightarrow \frac{\partial f}{\partial t}$. By assumption, $|d_n| \leq |\frac{\partial f}{\partial t}| \leq g(x)$ a.e. so by DCT,

$$\lim_{n \rightarrow \infty} \int d_n d\mu = \int \frac{\partial f}{\partial t}(x, t) d\mu.$$

Since $\int d_n d\mu = \int \frac{f(x, t+h_n) - f(x, t_o)}{h_n} = \frac{F(t) - F(t_o)}{h_n}$ then $F'(t) \rightarrow \int \frac{\partial f}{\partial t}(x, t) d\mu$. □

2. Prove that $\int_{(0, \infty)} x^n e^{-x} dx = n!$ by differentiating $\int_0^\infty e^{-tx} dt = 1/x$.

Proof. Starting with $\int_0^\infty e^{-tx} dt = \frac{1}{x}$, differentiate both sides such that $\frac{d^n}{dx^n} \frac{1}{x} = \frac{(-1)^n n!}{x^{n+1}}$ and $\frac{d^n}{dx^n} \int_0^\infty e^{-tx} dt = \int_0^\infty \frac{d^n}{dx^n} (e^{-tx}) dt = \int_0^\infty (-1)^n t^n e^{-tx} dt$. Then $\int_0^\infty t^n e^{-tx} dt = \frac{n!}{x^{n+1}}$ so when $x = 1$, $\int_0^\infty t^n e^{-tx} dt = n!$. If $x \in [a, b]$ then $t^n e^{-t} \leq t^n e^{-tx} \leq t^n e^{-ta}$ so the necessary conditions of theorem 1.129 are satisfied and the differentiation under the integral sign is valid. □

3. Prove that if f_n is a sequence of decreasing nonnegative measurable functions with $\int f_1 d\mu < \infty$, then

$$\int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Proof. Since $\int f_1 d\mu < \infty$ then $f_1 \in L^1(\mu)$ and f_1 dominates f_n as a sequence of decreasing nonnegative measurable functions. Furthermore, $\lim_{n \rightarrow \infty} f_n = 0$ since f_n is bounded below by 0 and decreasing. Thus DCT shows $\int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$. $\square$

4. Let $y_1, \dots, y_M$ be distinct points in $\mathbb{R}$, and $\beta_1, \dots, \beta_M \in \mathbb{R}$. Let

$$f(x) = \prod_{j=1}^M |x - y_j|^{-\beta_j}.$$

Find a necessary and sufficient condition on $(\beta_1, \dots, \beta_M)$ for f to belong to $L^1(\mathbb{R}, m)$.

Proof. The necessary and sufficient condition for $f \in L^1(\mathbb{R}, m)$ is that $\beta_j < 1$ for all $j \in \{1, 2, \dots, M\}$ and $\sum_{j=1}^M \beta_j < 1$ also. $\beta_j < 1$ is necessary to prevent $f(x)$ from diverging near y_j but this isn't sufficient because a small β_j like $\beta_j = \frac{1/2M}{\epsilon}$ would cause $\int f dm$ to diverge. Enforcing $\sum_{j=1}^M \beta_j < 1$ ensures $\int f dm$ is finite since $f(x) = |x - y_j|^{\beta_{product} < 1}$ whose integral does converge. $\square$

5. Define $g: \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} |x|^{-1/3}, & |x| < 1, \\ 0, & |x| > 1. \end{cases}$$

Let $\{r_k\}_{k=0}^\infty$ be an enumeration of all points in $[0, 1] \cap \mathbb{Q}$. Let

$$f(x) = \mathbf{1}_{[0,1]} \sum_{k \in \mathbb{N}} 2^{-k} g(x - r_k).$$

- (i) Show that $f \in L^1([0, 1])$.

Proof. Let $f_N(x) = \mathbf{1}_{[0,1]} \sum_{k=1}^N 2^{-k} g(x - r_k)$ be the N th partial sum. It follows that

$$\int_0^1 f(x) dx = \sum_{k=1}^N 2^{-k} \int_0^1 g(x) dx \tag{1}$$

$$= \sum_{k=1}^N \int_{-r_k}^0 (r_k - x)^{-1/3} dx + \int_0^{r_k} (x - r_k)^{-1/3} dx \tag{2}$$

$$= \frac{3}{2} r_k^{2/3} + \frac{3}{2} (1 - r_k)^{2/3} \leq 3. \tag{3}$$

Since $f_N(x) \rightarrow f(x)$ and $\int_0^1 f(x) dx$ is bounded then by MCT $\int_0^1 f(x) dx = \lim_{N \rightarrow \infty} \int_0^1 f_N(x) dx \leq 3$ so f is $\mathcal{L}^1([0, 1])$ and thus $L^1([0, 1])$ also. $\square$

- (ii) Show that f is unbounded on every open set (and remains so after modification on a set of measure zero). Show that f is discontinuous at every point in $[0, 1]$ and remains so after modification on a set of measure zero.

Proof. Let $O \subset [0, 1]$ be any subset of $[0, 1]$. Since $\mathbb{Q}$ is dense then $r_k \in O$ and when $x \neq r_k$, $g(x - r_k) = |x - r_k|^{-1/3}$. Notice that $f(x) \geq \frac{|x - r_k|^{-1/3}}{2^k} \rightarrow \infty$ as $r_k \rightarrow x$ so f is unbounded on O . Now let $h = f$ a.e. such that $E = \{x : h(x) \neq f(x)\}$ has $m(E) = 0$. For any $C > 0$

$$\{x \in O : f(x) > M\} \subseteq \{x \in O : f(x) > M\} \setminus M = \{x \in O : h(x) > M\}$$

which has non-zero measure, so $h(x)$ is also unbounded on O . Thus, f is unbounded on open intervals regardless of modification by a nullset. $\square$

- (iii) Let $0 < \varepsilon < 1$. Construct a compact subset K_ε of $[0, 1]$ such that $m(K_\varepsilon) \geq 1 - \varepsilon$ and $f\mathbf{1}_{K_\varepsilon}$ is continuous on K_ε .

Proof. Let $0 < \epsilon < 1$ and let $O_k = (r_k - \frac{\epsilon}{2^{k+2}}, r_k + \frac{\epsilon}{2^{k+2}})$. Observe that given $U = \cup_{k \in \mathbb{N}} O_k$, $m(U) \leq \sum_{k \in \mathbb{N}} m(O_k) = \frac{\epsilon}{2}$ by subadditivity. Define $K_\epsilon = [0, 1] \setminus U$ and observe that $m(K_\epsilon) \geq 1 - \frac{\epsilon}{2} \geq 1 - \epsilon$. Then for all $x \in K_\epsilon$, $|x - r_k| \geq \frac{\epsilon}{2^{k+1}}$ since $x \notin U$. Let $f_N(x) = \sum_{k=1}^N 2^{-k} g(x - r_k)$ and see

$$\sup |f(x) - f_N(x)| \leq \sum_{k=N+1}^{\infty} 2^{-k} \left(\frac{\epsilon}{2^{k+2}}\right)^{-1/3} = \frac{2^{2/3}}{\epsilon^{1/3}} \sum_{k=N+1}^{\infty} 2^{-2k/3} \rightarrow 0$$

by ratio test of geometric series. Thus f is uniformly continuous on K_ϵ . $\square$

6. Let $f \in L^1([0, 2\pi])$.

- (i) Prove that $\lim_{n \rightarrow \infty} \int_0^{2\pi} f(x) e^{-inx} dx = 0$.

Proof. First observe that $|e^{-inx}| \leq 1$. By theorem 1.117 let g be a step function such that $\int_0^{2\pi} |f - g| < \epsilon$. Then by the triangle inequality,

$$\left| \int_0^{2\pi} f e^{-inx} dx \right| \leq \left| \int_0^{2\pi} (f - g) e^{-inx} dx \right| + \left| \int_0^{2\pi} g e^{-inx} dx \right|.$$

Since g is a step function given sufficiently large n we know $\left| \int_0^{2\pi} g e^{-inx} dx \right| \rightarrow 0$. For the other piece, notice

$$\left| \int_0^{2\pi} (f - g) e^{-inx} dx \right| \leq \int_0^{2\pi} |f - g| dx < \epsilon.$$

Thus by substituting, $\int_0^{2\pi} f(x) e^{-inx} dx \leq \lim \left| \int_0^{2\pi} f(x) e^{-inx} dx \right| < \epsilon$ for sufficiently large n so $\lim_{n \rightarrow \infty} \int_0^{2\pi} f e^{-inx} dx = 0$. $\square$

(ii) Prove that for a set $E \subset [0, 2\pi]$ of positive Lebesgue measure $m(E)$,

$$\lim_{n \rightarrow \infty} \int_E \cos^2(nx + \gamma_n) dx = \frac{m(E)}{2},$$

for any sequence γ_n .

Proof. To begin use the trig identity for $\cos^2(\theta)$ such that $\cos^2(nx + \gamma_n) = \frac{1 + \cos(2nx + 2\gamma_n)}{2}$. Then $\int_E \cos^2(nx + \gamma_n) dx = \frac{m(E)}{2} + \frac{1}{2} \int_E \cos(2nx + 2\gamma_n) dx$. Using Euler's formula notice $\cos(2nx + 2\gamma_n) = \operatorname{Re}(e^{2inx} e^{2i\gamma_n})$ so by substitution $\int_E \cos(2nx + 2\gamma_n) dx = \operatorname{Re}(e^{2i\gamma_n} \int_E e^{2inx} dx) \leq |\int_E e^{2inx} dx|$ since $|e^{2i\gamma_n}| = 1$. By part i),

$$\lim_{n \rightarrow \infty} \int_E e^{2inx} dx = 0$$

so

$$\int_E \cos(2nx + 2\gamma_n) dx \rightarrow 0$$

since $E \subset [0, 2\pi]$. By substitution, $\lim_{n \rightarrow \infty} \int_E \cos^2(nx + \gamma_n) dx = \frac{m(E)}{2}$. $\square$

(iii) Let $F_n(x) = a_n \cos nx + b_n \sin nx$. Assume that the series $\sum_{n=0}^{\infty} F_n(x)$ converges on a set E of positive measure. Show that $\lim_{n \rightarrow \infty} a_n = 0$ and $\lim_{n \rightarrow \infty} b_n = 0$.

Hint: It might help to express $F_n(x)^2$ in the form $(r_n \cos(nx + \gamma_n))^2$, with $r_n > 0$.

Proof. Let $F_n(x)^2 = r_n^2 \cos^2(nx + \gamma_n)$ where $r_n^2 = a_n^2 + b_n^2$. We want to show $r_n \rightarrow 0$. Since $\sum_{n=0}^{\infty} F_n(x)$ converges on E then for all $x \in E$, $F_n(x) \rightarrow 0$ and furthermore $F_n(x)^2 \rightarrow 0$ pointwise also. Using Egorov's theorem, there exists a subset $A \subset E$ such that $F_n^2(x) \rightarrow 0$ uniformly on A which shows $\int_A F_n^2(x) dx \rightarrow 0$. Since $\int_A F_n(x)^2 dx = r_n^2 \int_A \cos^2(nx + \gamma_n) dx$ then

$$r_n^2 = \frac{\int_A F_n(x)^2 dx}{\int_A \cos^2(nx + \gamma_n) dx}.$$

Using the result part ii)

$$r_n^2 \rightarrow \frac{0}{m(A)/2} = 0$$

thus $a_n^2 + b_n^2 \rightarrow 0$ so individually $a_n \rightarrow 0$ and $b_n \rightarrow 0$. $\square$